LLM-Based QA from Structured Bibliographic Data via Semantic Similarity Understanding Network

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Abstract

In this work, we propose a new method for converting structured bibliographic data into a natural language-based Question Answering (QA) system built on a Large Language Model (LLM). The method, called Semantic Similarity Understanding Network (SSUN), combines a Transformer encoder for identifying semantic relationships between data entries with a text-to-text generation model for producing contextually appropriate answers. To evaluate its effectiveness, we conducted experiments by using paraphrased queries and compared SSUN with several baseline models, including T5-Base, T5-Small, BART, and ProphetNet. The results indicate that SSUN achieves higher accuracy and robustness, particularly in handling queries that are semantically equivalent but phrased differently. Notably, SSUN can also detect and appropriately respond to out-of-domain questions by returning "no relevant information" instead of generating inaccurate results. These findings suggest that SSUN provides a more flexible and semantically aware interface for querying structured bibliographic data through natural language.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Natural language processing; Natural language generation.

Keywords

Semantic Similarity, Understanding, Structured Bibliographic Data, LLM, QA

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1 Introduction

In modern library systems, bibliographic data is typically stored in structured forms such as relational databases or spreadsheets. These formats are efficient for managing and organizing large-scale information, but they are not designed for interaction through natural language. Users are generally required to understand the data schema and write queries using formal languages like SQL. For those without technical training, this presents a significant obstacle, often making the process of data access complex, time-consuming, and less approachable to general users.

Recent progress in Natural Language Processing (NLP) [1, 2] and the advancement of Large Language Models (LLMs) has introduced promising alternatives for accessing structured data. Models like T5 [3, 4] have shown strong capabilities in interpreting user questions written in natural language and returning meaningful answers. These models remove the need for specialized query knowledge, making interaction with complex datasets more intuitive. This paradigm shift moves beyond traditional systems and opens the possibility for a broader range of users to access information through conversational input.

Nevertheless, conventional systems still rely heavily on fixed query structures and often fail to respond accurately when questions are reworded or when underlying data formats change. This restricts their flexibility and limits their usefulness in real-world, dynamic environments. By contrast, LLM-based approaches using semantic similarity can better handle linguistic variation and connect diverse query formulations with relevant results. This makes them more robust and adaptable, especially for users who may express queries in informal or unexpected ways.

In this work, we propose a method for converting structured bibliographic data into a natural language-based Question Answering (QA) system [5-8] using a Large Language Model. The proposed method, called Semantic Similarity Understanding Network (SSUN), is capable of aligning meaningfully similar user queries with relevant data records, regardless of surface-level phrasing. By combining semantic encoding [9, 10] through a Transformer with the T5 model for answer generation, SSUN improves both accuracy and flexibility in querying bibliographic resources. The system lowers the barrier to access and offers a more human-like interface for library data exploration.

The main contributions of this paper are as follows:

1. A method is presented to transform structured bibliographic data into knowledge graph data, laying the foundation for generating relevant training samples. These samples are then used to train the model, enabling it to process and respond to natural language queries accurately.
2. Inspired by the Transformer architecture, a Semantic Similarity Understanding Network (SSUN) is proposed, which can enhance the T5 model. This network utilizes an encoding method to compute semantic similarity between texts. It enables the transformation of structured data into appropriate

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Table 1: Structured bibliographic data

Title	Author	Type	Field	Year	Location	Status
Radium	Marie Curie	Book	Chemistry	1898	Science Section A1	Active
Theory of Relativity	Albert Einstein	Book	Physics	1905	Science Section A2	Active
Moons of Jupiter	Galileo Galilei	Book	Astronomy	1610	Astronomy Section B1	Active
Theory of Evolution	Charles Darwin	Book	Biology	1859	Biology Section C3	Active
AC Electricity	Nikola Tesla	Book	Engineering	1888	Engineering Section D1	Active
Light Bulb	Thomas Edison	Book	Engineering	1879	Engineering Section D2	Active
Gravity	Isaac Newton	Book	Physics	1687	Science Section A3	Active
Turing Machine	Alan Turing	Book	Computer Sci	1936	Computing Section E1	Active
Black Hole Radiation	Stephen Hawking	Book	Physics	1974	Science Section A4	Active

training inputs. This allows for accurate querying and the identification of semantically similar sentences, improving the overall query process.

2 Methodology

2.1 2.1 Transforming Structured Data into Training Samples

In recent years, there has been growing attention on converting structured bibliographic data into formats that are more suitable for Natural Language Processing (NLP) applications. This paper proposes a method for transforming structured bibliographic records into a knowledge graph [11-13], which enhances the system's semantic understanding and enables more effective querying using natural language. The overall process involves extracting key entities and the relationships between them from structured data, building a corresponding knowledge graph, and generating training samples for a Question Answering (QA) system based on a Large Language Model (LLM). Table 1 displays an example of the original data presented in tabular form, and Figure 1 shows the resulting knowledge graph constructed from that data.

The transformation begins by analyzing the structured data—such as bibliographic tables—to extract meaningful entities (e.g., scientists, inventions) along with their semantic relationships (e.g., "discovered," "developed"). These components are then used to build a knowledge graph that captures the underlying structure and meaning of the data in a more interconnected way. This graph is subsequently used to generate training data that helps the LLM-based QA model better interpret user questions and provide accurate, context-aware responses to natural language queries.

These question-answer pairs are generated by converting structured data into a knowledge graph, where scientists and discoveries are nodes, and relationships (e.g., "discovered," "developed") are the edges connecting them. Figure 2 below shows the resulting knowledge graph, where entities (scientists) are connected by relationships (edges) to their discoveries.

The knowledge graph is then used as training data for the LLM-based QA system, improving its ability to respond to natural language queries. This transformation allows the system to match semantically similar queries, even if the phrasing is different. For example, when the data for Marie Curie, her discovery of Radium, and the relationship "discovered" are extracted, a question-answer pair can be generated as follows:

- The question generated from the data is: "What did Marie Curie discover?"
- The corresponding answer is: "Marie Curie discovered Radium."

This question-answer pair serves as a training sample for the LLM-based Question Answering (QA) system. The model learns to associate the scientist (Marie Curie) with the discovery (Radium) through the relationship ("discovered"). Such pairs are used to train the system, enabling it to accurately respond to natural language queries by generating relevant answers based on the structured bibliographic data.

2.2 2.2 Semantic Similarity Understanding Network (SSUN)

The Semantic Similarity Understanding Network (SSUN) improves question-answering performance by focusing on semantic similarity, which helps match queries with relevant data, as demonstrated in Figure 2. The process begins by tokenizing the input query and adding token and position embeddings. Sentence embeddings are then created using a Sentence Transformer, capturing the semantic meaning of both the query and training data. These embeddings pass through self-attention layers in the Transformer encoder, allowing the model to highlight the most relevant parts of the sentence and comprehend the context effectively.

Once the embeddings are obtained, the semantic similarity between the query and training samples is computed using cosine similarity. Based on this, the most similar training data is selected. The model then generates an answer by decoding the corresponding training sample. This ensures that the system delivers accurate responses, even when the queries are semantically similar or paraphrased. By fine-tuning the system with question-answer pairs from a knowledge graph, SSUN enhances its ability to process queries and generate contextually accurate responses.

The following formulas outline the core computations in Figure 2 for both the encoder and decoder of the Transformer model. They describe how the model processes input data using attention mechanisms, computes contextual embeddings, and generates output tokens. These computations are essential for tasks such as

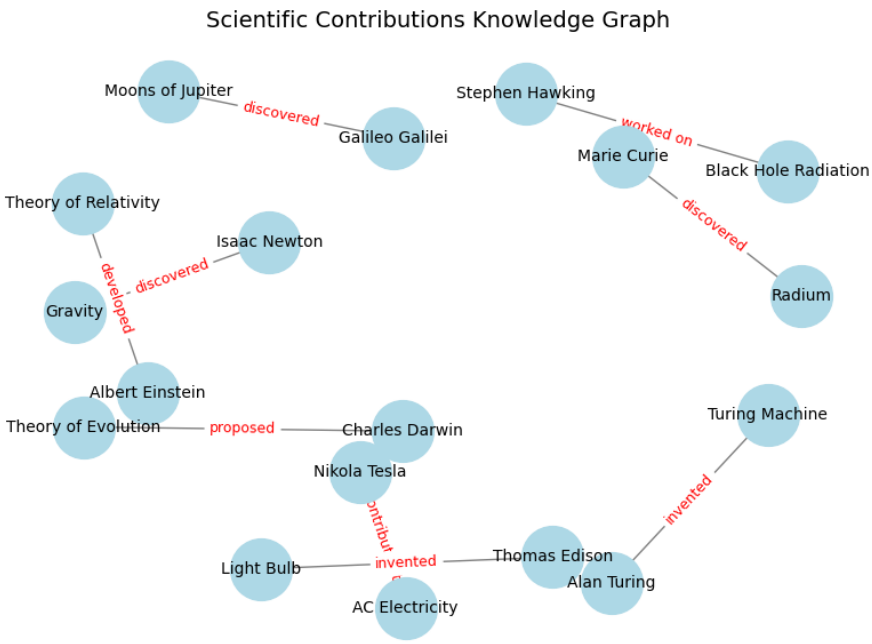

Figure 1: Knowledge graph representation of bibliographic data

semantic similarity matching and answer generation in the SSUN architecture.

Self-Attention:

$$\text{Attention}\,(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

Multi-Head Attention:

$$\text{Multi} - \text{Head Attention}\,(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \tag{2}$$

Feed-Forward Network (FFN):

$$\text{FFN}\,(x) = \max\,(0, xW_1 + b_1)\,W_2 + b_2 \tag{3}$$

Masked Attention (in Decoder):

$$\text{Masked Attention}\,(Q, K, V) = \text{softmax}\left(\frac{QK^T + M}{\sqrt{d_k}}\right)V \tag{4}$$

Encoder-Decoder Attention:

$$\text{Encoder} - \text{Decoder Attention}\,(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{5}$$

Output Generation:

$$\text{Output Tokens} = \text{softmax}\,(W_o \cdot \text{Decoder Output}) \tag{6}$$

The Transformer model uses several key parameters across its mechanisms. In Self-Attention, Q (query), K (key), and V (value) vectors are used with the formula Attention(Q, K, V) = softmax $(\frac{QK^T}{\sqrt{d_k}})V$. Multi-Head Attention combines multiple attention heads, Multi-Head Attention(Q, K, V) = Concat(head1, . . . , headh) W_o where W_o is the output matrix. FFN applies FFN(x) = $\max(0, xW_1 + b_1)W_2 + b_2$, with weight matrices $W1$, $W2$ and biases $b1$, $b2$. In Masked Attention for the decoder, Masked Attention(Q, K, V) = softmax $(\frac{QK^T+M}{\sqrt{d_k}})V$, where M is the mask matrix. Encoder-Decoder Attention uses Encoder-Decoder Attention(Q, K, V) = softmax $(\frac{QK^T}{\sqrt{d_k}})V$. Finally, Output Generation is Output Tokens = softmax($W_o \cdot$ Decoder Output) , with W_o as the output weight matrix. These parameters and operations allow the Transformer to effectively process and generate language.

2.3 2.3 Training Process of SSUN

The training process of the Semantic Similarity Understanding Network (SSUN) includes several crucial steps designed to improve question-answering performance. It begins by preparing training data as question-answer pairs derived from structured bibliographic data. These pairs are tokenized and formatted to be used by the model.

Step 1: Semantic Encoding – The training data is converted into semantic vectors using Sentence Transformers. This process creates a representation that captures the meaning behind the text, enabling the model to understand the relationships between queries and answers.

Step 2: Semantic Similarity Calculation – The input query is compared with the encoded training data using cosine similarity, helping the model identify the most semantically relevant training example.

Step 3: Selection of Relevant Training Data – Based on the similarity score, the most relevant training example is selected, ensuring the query is matched with a closely related training sample.

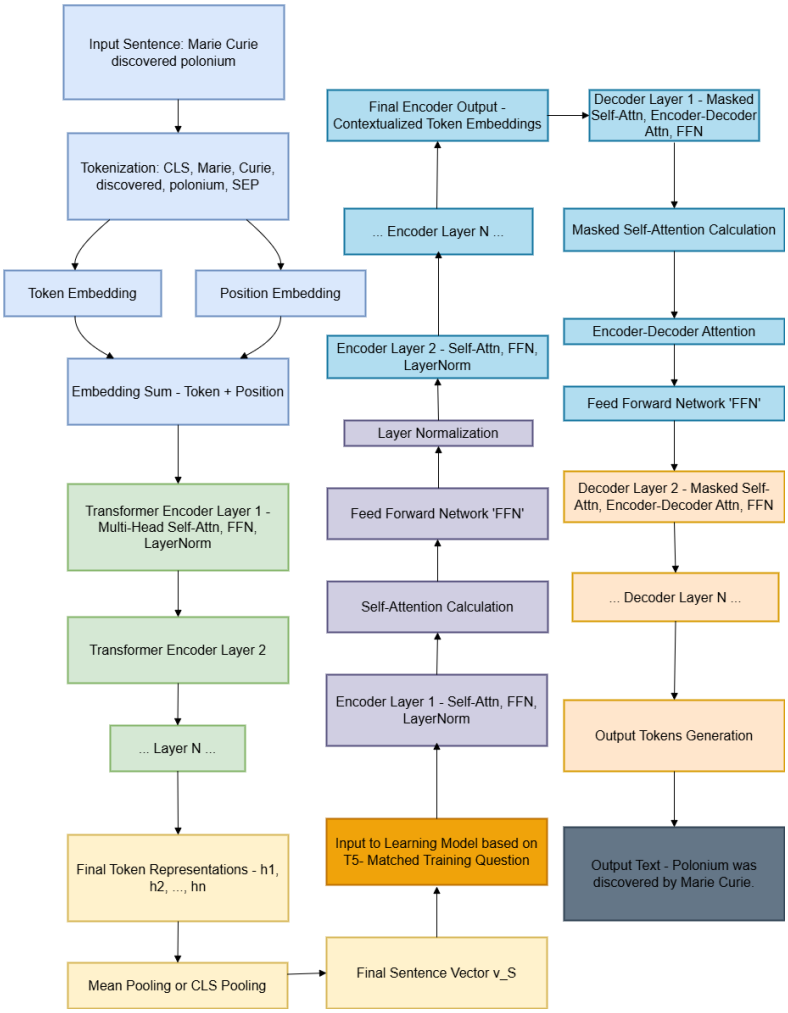

Figure 2: Framework of semantic similarity understanding network (SSUN)

Step 4: Answer Generation – The selected training sample is processed by the trained model, which generates an answer based on the query and the relevant training data.

3 Experiment and Evaluation

The experiment tests the SSUN with paraphrased questions to evaluate its ability to match queries with relevant training data, as shown in Table 2. Each query is compared with the training data using semantic similarity, with cosine similarity scores ranging from 0.8590 to 0.9421. The model successfully identifies the most similar training question and generates accurate answers. For example, when the query is "What did Marie Curie discover?", the predicted answer is "Marie Curie discovered radium," which matches the standard response.

However, when the query concerns a topic outside the scope of the training data, such as "What's the capital of Mars?", the model correctly identifies that the training samples do not contain relevant information. Since the structured data does not include such a query, the model responds appropriately with "The database needs to be improved, there is no relevant information." This outcome shows that while SSUN excels at answering questions within the scope of its training, it cannot handle out-of-scope queries due to the limitations of the training data.

In the next experiment, five models—T5-Base [3, 4], T5-Small [3, 4], Bart [14], ProphetNet [15], and SSUN—are evaluated for their ability to generate accurate responses to paraphrased natural language questions. The evaluation is conducted using a standardized set of semantically rephrased queries derived from the original training context, and the results are reported in Table 3. An accuracy rate of 70% is recorded for T5-Base, indicating that most reworded inputs are correctly processed when their phrasing remains closely aligned with the training distribution. Nevertheless, performance degradation is observed when the model is exposed to queries that diverge significantly in structure or domain, leading to occasional generation of vague or irrelevant content.

For T5-Small, a lower accuracy of 50% is observed. While certain paraphrases are correctly recognized and handled, reduced model capacity appears to limit its generalization ability. Precision tends

Table 2: Testing Results of SSUN for Paraphrased Question Matching

Input Question	Most Similar Training Question	Similarity Score	Predicted Answer	Matched Standard Answer	Result
What did Marie Curie discover?	What did Marie Curie discover?	0.9421	Marie Curie discovered radium.	Marie Curie discovered radium.	Correct
What was discovered by Marie Curie?	What did Marie Curie discover?	0.9382	Marie Curie discovered radium.	Marie Curie discovered radium.	Correct
Can you tell me what Marie Curie discovered?	What did Marie Curie discover?	0.9340	Marie Curie discovered radium.	Marie Curie discovered radium.	Correct
What is Marie Curie famous for discovering?	What did Marie Curie discover?	0.8590	Marie Curie discovered radium.	Marie Curie discovered radium.	Correct
Which discovery is attributed to Marie Curie?	What did Marie Curie discover?	0.8956	Marie Curie discovered radium.	Marie Curie discovered radium.	Correct
What did Marie Curie find?	What did Marie Curie discover?	0.9114	Marie Curie discovered radium.	Marie Curie discovered radium.	Correct
What's the capital of Mars?	Who discovered moons of Jupiter?	0.2862	The database needs to be improved, there is no relevant information	N/A	Incorrect (No relevant info)

to degrade when handling more complex or indirect paraphrases, often resulting in semantic ambiguity.

BART also achieves 50% accuracy, though its outputs are often characterized by excessive verbosity and content hallucination. Incorrect associations, such as attributing astronomical discoveries to irrelevant historical figures or introducing fictitious facts, are repeatedly generated in response to both in-domain and out-of-domain questions.

A markedly lower performance is exhibited by ProphetNet, for which 0% accuracy is reported. Across all tested inputs, the model consistently fails to return correct answers. Instead, its outputs are dominated by lengthy, speculative, and factually incorrect statements, suggesting an inability to generalize beyond surface-level text patterns.

By contrast, SSUN yields a perfect 100% accuracy. All semantically paraphrased questions are accurately answered, and unrelated or out-of-domain inputs are reliably identified and responded to with "No relevant information." This performance can be attributed to SSUN's architectural emphasis on semantic similarity modeling, which enables robust alignment between meaning-preserving rephrasings and relevant knowledge.

Overall, these findings demonstrate that SSUN substantially outperforms traditional generative models under paraphrase-aware evaluation. Its ability to avoid hallucinated outputs and handle semantic variation with precision highlights its potential for deployment in practical, high-stakes question answering applications.

4 Conclusion

This paper presents the Semantic Similarity Understanding Network (SSUN), a method designed to convert structured bibliographic data into an effective Question Answering (QA) system built on Large Language Models (LLMs). By combining semantic similarity matching with the T5 model for answer generation, SSUN is able to produce accurate responses, even when users phrase their queries differently. Experimental evaluations show that SSUN outperforms conventional querying methods, particularly in cases where paraphrased or loosely structured questions are used. The approach proves especially useful for retrieving information from knowledge graphs and other structured data sources.

SSUN relies on several core computational components, including Sentence Transformer embeddings, self-attention mechanisms, and a Transformer-based encoder-decoder architecture. These components work together to help the model understand both the input queries and the training data. Cosine similarity is used to match incoming questions with the most relevant training samples, improving the model's ability to select contextually appropriate information and generate accurate answers. Due to this design, the system offers a scalable solution for knowledge retrieval, with potential applications in digital library systems, academic data analysis, and broader content management tasks.

That said, SSUN is not without limitations. Its performance is highly dependent on the quality and diversity of the training data. If the dataset lacks coverage in certain areas, the model may struggle with out-of-domain queries or unfamiliar topics. Although it performs well with semantically similar input, it can still be challenged by vague, contradictory, or highly ambiguous questions. Additionally, applying SSUN to large-scale datasets may result in higher computational costs, both during training and inference.

Looking ahead, future improvements will aim to expand the system's training data sources, enhance its performance in practical environments, and extend its capabilities to handle more complex and nuanced queries. Overall, the results indicate that SSUN is a promising step forward in the field of automated knowledge retrieval and natural language understanding [16, 17].

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Table 3: Comparative experiments across methods

Input Question	T5-Base	T5-Small	Bart	ProphetNet	SUUN
What did Marie Curie discover?	Marie Curie discovered radium.	Marie Curie discovered radium.	Marie Curie discovered radium.	marie marie marie marie cu moons early cu marie moons marie cu marie moons marie marie cu marie marie marie moons marie early marie gal gal marie the the marie ada ada marie of of	Marie Curie discovered radium.
What was discovered by Marie Curie?	Marie Curie discovered radium.	Marie Curie discovered radium.	Marie Curie discovered radium.	marie marie marie marie marie moons marie moons marie marie moons moons marie cu marie marie cu marie cu marie early marie ra marie marie early marie marie ra marie early cu ra early marie ra marie the marie marie of marie	Marie Curie discovered radium.
Can you tell me what Marie Curie discovered?	Marie Curie discovered radium.	Marie Curie discovered radium.	Marie Curie discovered radium.	marie marie cu moons cu marie marie cu marie moons marie marie moons moons cu cu moons marie early the ra early cu cu moons moons cu early marie the cu the marie love	Marie Curie discovered radium.
What is Marie Curie famous for discovering?	Marie Curie discovered radium.	Marie Curie is famous for discovering.	Marie Curie is known for early computer programming	marie for for for marie for for for for marie for marie for marie marie for early for for moons for for early marie marie marie early for moons for marie early for marie moons for early for moons moons for early marie for moons marie	Marie Curie discovered radium.
Which discovery is attributed to Marie Curie?	Marie Curie discovered radium.	Marie Curie discovered radium.	Marie Curie is attributed to early computer programming.	marie discovery moons is of early the for ra cu ada galileo. jupiter computering love gal ac	Marie Curie discovered radium.

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What did Marie Curie find?	Marie Curie discovered radium.	Marie Curie found radium.	Marie Curie discovered radium.	marie marie marie marie marie cu marie marie cu marie marie early marie cu cu marie early marie marie moons marie marie ada marie marie gal marie marie the marie marie of marie marie for marie marie ra marie marie ac marie marie. marie marie jupiter marie marie love marie marie juno marie	Marie Curie discovered radium.
What's the capital of Mars? (not the training sample)	Mars is the capital of Mars.	Mars is the capital of Mars.	Galileo Galilei discovered moons of Jupiter.	the moons ada of early for jupiter computer ac gal is. ra on earth	No relevant information.
What was the breakthrough that Marie Curie made?	Marie Curie made the breakthrough that Marie Curie made radium.	Marie Curie made the breakthrough.	Marie Curie made the theory of radium.	arie marie marie marie the marie the early cu the marie ada ada marie early marie moons for marie cu marie early marie ada marie for ac moons marie cu marie for marie	Marie Curie discovered radium.
What did Marie Curie contribute to science?	Marie Curie contributed to science.	Marie Curie contributed to science.	Marie Curie contributed to AC electricity.	marie early science jupiter moons the ra gal juno cu for of computer ada galileo to. ac	Marie Curie discovered radium.
Can you tell me the major discovery made by Marie Curie?	Marie Curie discovered radium.	Marie Curie discovered the major discovery.	Marie Curie made the theory of radium.	marie marie marie marie the cu moons for ra early of the marie cu marie moons marie for marie ra marie early marie of marie love love marie ada ada marie	Marie Curie discovered radium.
Accuracy	70%	50%	50%	0%	100.00%

Correct Incorrect

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